

THE ROLE OF CONVECTION IN THE EXISTENCE OF WAVEFRONTS FOR POPULATION DYNAMICS WITH BIAS IN MOVEMENTS

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Torino

We deal with the following equation:

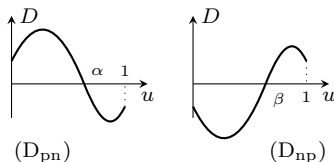
$$u_t + f(u)_x = \{D(u) u_x\}_x + g(u) \quad \text{for } x \in \mathbb{R}, t \geq 0. \quad (1)$$

Here, the unknown is

$$u = u(x, t) \in [0, 1],$$

while f , D and g are given and such that

- the **diffusivity** D is possibly degenerate ($D(u_0) = 0$ for some u_0) or even negative in some interval;
- the **source** g is either identically 0 or vanishes at some isolated points (equilibria);
- no assumptions on the sign **convection** f . For convenience, $f(0) = 0$.

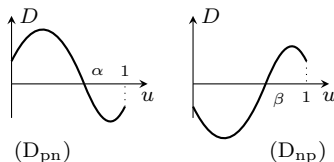


Equation (1) is then

- **forward** parabolic equation where $D > 0$;
- **backward** parabolic equation where $D < 0$;
- hyperbolic equation where $D = 0$.

D can be **negative**. This occurs in the context of:

- *collective movements* (vehicular flows and crowd dynamics): $D < 0$ near to 1, the negativity of D simulates an **aggregative behavior under high density of the agents**:
 - high car densities and limited sight distance ahead in *vehicular flows* (Nelson (2000));
 - panic behavior in *overcrowded dynamics* (Colombo-Rosini (2005) or Corli-Malaguti (2019)).
- several *physical* and *biological* models.



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Traveling wave solutions (TWs for short) are solutions u which takes the form

$$u(x, t) = \varphi(x - ct),$$

for some wave profile $\varphi = \varphi(\xi)$ and a wave speed $c \in \mathbb{R}$.

We then formally rephrase (1) as

$$\left(D(\varphi) \varphi' \right)' + \left(c - f'(\varphi) \right) \varphi' + g(\varphi) = 0. \quad (2)$$

TW of (1) if there exist $\varphi \in C(I)$, for $I \subset \mathbb{R}$, φ differentiable a.e. in I and c such that φ satisfies (2) in distributional sense in I .

-) $\varphi \in C(I)$ is a function of class $C(I)$, $I \subset \mathbb{R}$, with values in $[0, 1]$, differentiable a.e. and such that $D(\varphi) \varphi' \in L^1_{\text{loc}}(I)$.
-) for every $\psi \in C_0^\infty(I)$,

$$\int_I \left(D(\varphi(\xi)) \varphi'(\xi) - f(\varphi(\xi)) + c\varphi(\xi) \right) \psi'(\xi) - g(\varphi(\xi)) \psi(\xi) d\xi = 0.$$

A bounded TW, whose profile is defined globally ($I = \mathbb{R}$), non-constant and **monotone**, and

$$\varphi(-\infty) = \ell^+ \quad \varphi(+\infty) = \ell^-,$$

is said to be a **wavefront** (WF) from ℓ^+ to ℓ^- , where $g(\ell^\pm) = 0$.

$\mathcal{J}$ is the set of **admissible speeds**, that is the set of $c \in \mathbb{R}$ such that the considered problem admits a WF with speed c .

Why WFs?

- Wave phenomena are observed in many natural reaction, convection and diffusion processes;
- such solutions can be used as **tools in comparison principles** and the like to **determine the properties** of general solutions;
- they are **attractors** of more general solutions for wide classes of initial data;

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WHY WFs? (LONG-TERM BEHAVIOUR)

- in conformance with their natural occurrence in many mathematically modelled phenomena, **they characterize the long-term behaviour** in numerous situations. Assume then

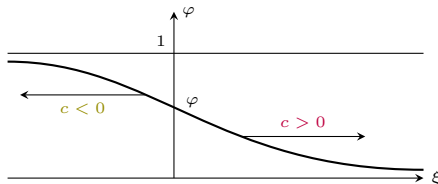
$$\varphi(-\infty) = 1, \quad \varphi(\infty) = 0$$

-) $c > 0$: **survival** and the population fills the entire space;

$$\lim_{t \rightarrow +\infty} u(x, t) = 1$$

-) $c < 0$: **extinction** of the population.

$$\lim_{t \rightarrow +\infty} u(x, t) = 0$$



In this talk, we discuss the **existence, uniqueness (non-uniqueness) and regularity** (where D degenerates) of WFs, **highlighting how the convection f affects these results.**

Key issues are also:

- Information on $\mathcal{J}$.
- The occurrence of profiles which reach an equilibrium for a **finite value ξ^*** .

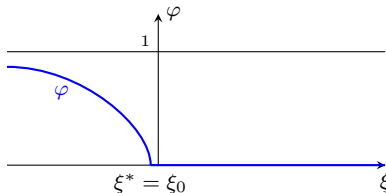


FIGURE: Non-regular profile at ξ^* .

A few (old) references on (biological) motivations and solutions to (E) in the case $D > 0$:

-) Fisher (1937)
-) Kolmogorov-Petrovskii-Piskunov (1937)
-) Aronson-Weinberger (1978)
-) Fife (1979)

Negative diffusivity Several models use equation (E) in the case D is negative somewhere:

Turchin (1989) *Aggregation in biology*, Barenblatt-Bertsch-Dal Passo-Ughi (1993) *Oceanography*, Kerner-Osipov (1994) *Thermodynamics*, Nelson (2000) *Instabilities in traffic flows*, DiCarlo-Juanes-Tara-Witelski (2008) *Geophysical flows*, Bruno-Tosin-Tricerri-Venuti (2011) *Pedestrian flows*.

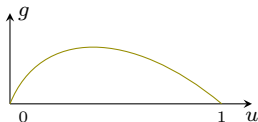
Few general results, for instance:

Plotnikov (1993–1996), Evans-Portilheiro (2004) (equation is generally ill-posed $\rightarrow$ viscous regularization through $\dots + \varepsilon \Delta u_t^\varepsilon$)

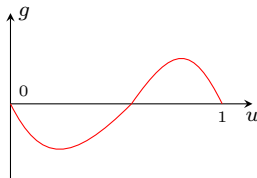
Bertsch, Mascia, Smarrazzo, Terracina, Tesei (2007–).

In this talk:

-) $g \in C^0([0, 1])$, $g(0) = g(1) = 0$ and either g is **monostable** or **bistable** (Allen-Cahn):



g monostable



g bistable

We consider WFs s.t.

$$\boxed{\varphi(-\infty) = 1, \quad \varphi(\infty) = 0} \quad (3)$$

Typical results in the case $f = 0$

-) if g is **monostable**, WFs if and only if $c \geq c^* = c^*(Dg)$, and the corresponding profile is unique (up to space shifts);
-) if g is **bistable**, WFs for **at most** a unique value of wave speed c^* , and the corresponding profile is unique (up to space shifts).
-) Moreover, profiles are smooth except (possibly) at 0, 1.

Take $D \geq 0$ and g monostable. In Marcelli-Malaguti (2004), improved in B.-Corli-Malaguti (2020) (see also Marcelli-Papalini (2018)):

WFs for $\mathcal{J} = [c^*, +\infty)$, with $c^* = c^*(Dg, f)$ such that

$$\min \left\{ \sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi}, f'(0) + 2\sqrt{\liminf_{\varphi \rightarrow 0^+} \frac{D(\varphi)g(\varphi)}{\varphi}} \right\} \leq c^* \\ \leq 2\sqrt{\sup_{(0,1]} \frac{D(\varphi)g(\varphi)}{\varphi}} + \sup_{(0,1]} \frac{f(\varphi)}{\varphi}.$$

-) c_f comes from the diffusion-convection equation of the type $u_t + f(u)_x = u_{xx}$. Gilding-Kersner (2004): WFs from $\ell \in (0, 1]$ to 0 iff

$$\frac{f(\ell)}{\ell} \varphi > f(\varphi) \quad \text{for } \varphi \in (0, \ell). \quad (C)$$

Hence, c_f is the maximal speed of TWs from $\ell \in (0, 1]$ to 0.

-) Condition (C) is also necessary and sufficient condition in order that

$$u_t + f(u)_x = 0$$

admits a shock wave of speed $f(\ell)/\ell$.

Take $D \geq 0$ and g monostable. In Marcelli-Malaguti (2004), improved in B.-Corli-Malaguti (2020) (see also Marcelli-Papalini (2018)):

$(Dg)'(0) = 0 \rightarrow$ WFs for $\mathcal{J} = [c^*, +\infty)$, with $c^* = c^*(Dg, f)$ is such that

$$\sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi} \leq c^* \leq 2 \sqrt{\sup_{\varphi \in (0,1]} \frac{D(\varphi)g(\varphi)}{\varphi}} + \sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi}.$$

that reads

$$c_f \leq c^* \leq c_{Dg} + c_f$$

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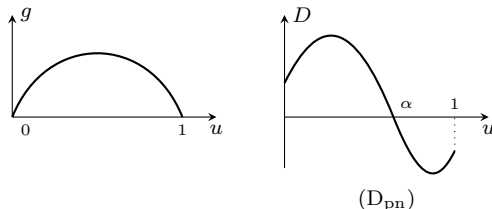
Hence, c_f is the *maximal speed* of TWs from $\ell \in (0, 1]$ to 0.

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The case g monostable and (D_{pn})



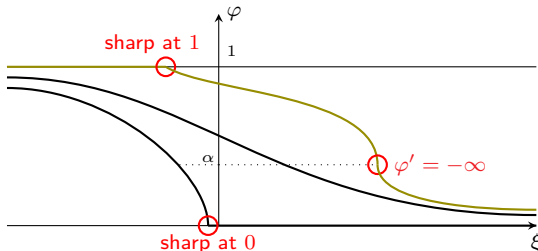
Theorem [B.-Corli-Malaguti (2021)]

- $\mathcal{J} = [c_{pn}^*, +\infty)$, with estimates on c_{pn}^* depending on Dg and f , that are coherent with the case $f = 0$;
- For every admissible speed, there is a **unique** wavefront solution (up to space shifts);
- precise estimates on c_{pn}^{*1} ;

¹[B., Corli, Malaguti (2020)]

Sharp profiles

- at 0 or 1 when $c = c_{pn}^*$; at α also for $c > c_{pn}^*$;
- f nonlinear $\Rightarrow$ the occurrence of *infinity-slope* profiles



Further results with g monostable

- negative-to-positive (D_{np});
- two sign-changing (D_{pnp}), (D_{nnp});

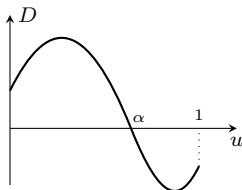
When does a profile become vertical?

-) f needs to **change convexity from convex to concave**:

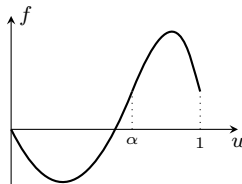
(because of $c > f'(\alpha)$, $c \geq c_{pn}^* = c_{pn}^*(Dg, f)$: these conditions fails if f is concave or convex);

-) roughly speaking, the behavior of f strongly contrasts that of D ;

i.e., the **diffusion rate D' must be opposite to the convective speed f'** .

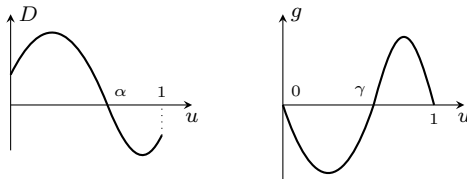


(D_{pn})



A THIRD SURPRISING EFFECT OF CONVECTION: EXISTENCE AND NONUNIQUENESS OF SOLUTIONS

Forward-backward equations with bistable reaction term



$f = 0$: in [Maini-Malaguti-Marcelli-Matucci (2007)] was considered.

$$\alpha > \gamma, \quad c > 0 \quad \text{and} \quad \int_0^\alpha D(u)g(u) \, du > 0$$

are needed in order to have solutions.

At most, a **unique admissible** $c^* > 0$ for which (1) admits a (unique up to shifts) wavefront.

THEOREM 1 (ADMISSIBLE SPEEDS, [B.-CORLI-MALAGUTI 2022])

Either $\mathcal{J}$ is empty or

–) $\alpha > \gamma$: then

$\mathcal{J}$ contains a unique value

▷ the corresponding profile is unique (up to shifts);

–) $\alpha = \gamma$: then

$\mathcal{J}$ is a bounded interval

▷ for $c \in \mathcal{J}$ the profile is unique up to shifts and **stretchings** at level γ .

▷ Moreover, if it holds

$$|g(\varphi)| \geq L|\varphi - \gamma|^\tau \quad \tau \in (0, 1), \quad \text{in a neighborhood of } \gamma, \quad (*)$$

then $\mathcal{J}$ is also closed;

–) $\alpha < \gamma$: then

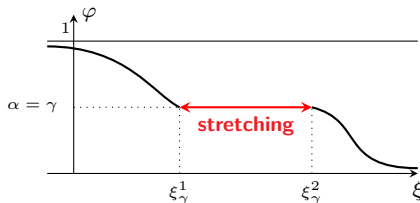
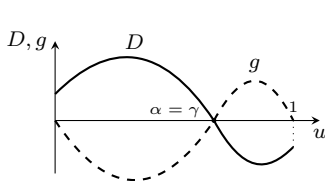
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▷ for $c \in \text{int} \mathcal{J}$ there is a family of profiles $\{\varphi_\lambda\}_\lambda$;

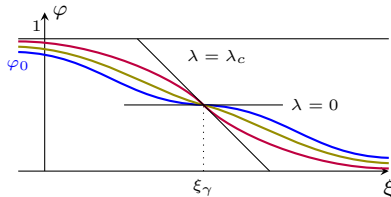
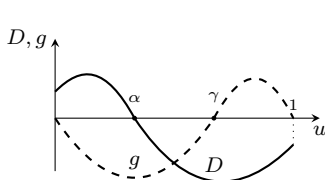
▷ Every φ_λ is unique (up to shifts) such that $\varphi'_\lambda(\xi_\gamma) = \lambda \in [\lambda_c, 0)$;

▷ If $(*)$ holds, then $\mathcal{J}$ is also closed and for $c \in \mathcal{J}$ there exist profiles corresponding to $\lambda = 0$; φ_0 is unique up to shifts and **stretchings** at level γ .

- All cases actually occur.
- $\alpha = \gamma$: uniqueness up to stretchings at γ



- $\alpha < \gamma$: essential lost of uniqueness



- Plateaus at γ are possible if $D(\varphi)\varphi'$ vanishes as φ reaches γ : this is never the case if $\alpha > \gamma$.

- In case $\alpha \leq \gamma$, the set $\mathcal{J}$ **can degenerate** to a single value.
- There is no continuity of results when $\gamma \rightarrow 0$: in the case $\gamma = 0$ there is a whole half line of admissible speeds [B.-Corli-Malaguti (2021)].
- Why the loss of uniqueness if $\alpha < \gamma$? This phenomenon, discovered in B.-Corli-Malaguti (2020) does **not** depend on the negativity of D .
- **Precise estimates** on $\inf \mathcal{J}$ and $\sup \mathcal{J}$ can be given.

[–] The **lack of regularity** properties of the involved functions $\implies$ it is **not** possible to apply a standard dynamical system analysis, in order to solve (2)-(3).

[–] Due to the monotonicity of φ , we instead take advantage of a method well-formalized by **Marcelli-Malaguti** in the early '00s: **reduction** (in regions where $D \neq 0$) of (2)-(3) into the following singular first-order boundary problem

$$\begin{cases} \dot{z}(\varphi) = f'(\varphi) - c - \frac{D(\varphi)g(\varphi)}{z(\varphi)} & \text{for } \varphi \in (0, \alpha) \cup (\alpha, 1), \\ z(\varphi) < 0 & \text{for } \varphi \in (0, \alpha), \\ z(\varphi) > 0 & \text{for } \varphi \in (\alpha, 1), \\ z(0) = z(\alpha) = z(1) = 0. \end{cases} \quad (4)$$

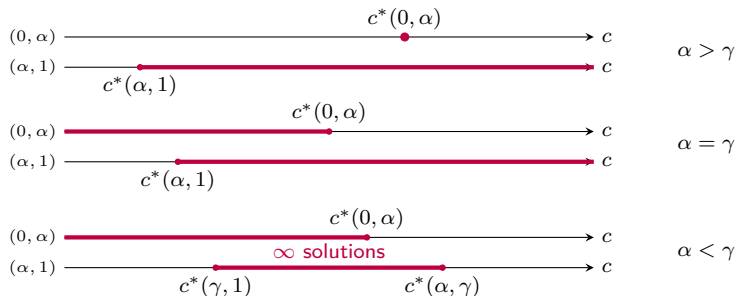
[–] The reduction is essentially formalized by the definition of the function $z = z(\varphi)$ as

$$z(\varphi) := D(\varphi)\varphi', \quad (5)$$

where the r.h.s. is evaluated at $\varphi^{-1}(\varphi)$.

[–] We study the solvability of (4) essentially by means of a **refined use of upper- and lower-solutions** based on **comparison-type techniques**.

[–] Depending on the mutual order between α and γ it is possible to solve (4).



[–] The presence of f implies that the **intervals in which the sub-problems have a solutions** can **overlap**.

[–] The presence of a family $\{\varphi_\lambda\}_\lambda$ depends upon the fact that in the case $\alpha < \gamma$, for any admissible speed there exists a **family of solutions** z of (4), parameterized by the value $z(\gamma)$.

THEOREM 2 (SUFFICIENT CONDITIONS [B.-CORLI-MALAGUTI 2022])

Assume that f is strictly concave. Then, WFs if

(A) $\alpha > \gamma$, $\dot{g}(\gamma) > 0$ and

$$\dot{f}(\gamma) - \frac{f(1) - f(\alpha)}{1 - \alpha} \geq 2 \left(\sqrt{\sup_{[\alpha, 1)} \delta(Dg, 1)} + \sqrt{\sup_{[0, \gamma)} \delta(Dg, \gamma)} \right); \quad (6)$$

(B) $\alpha = \gamma$, $(*)$ and (6);

(C) $\alpha < \gamma$ and

$$\dot{f}(\gamma) - \frac{f(1) - f(\gamma)}{1 - \gamma} > 2 \left(\sqrt{\sup_{[\gamma, 1)} \delta(Dg, 1)} + \sqrt{\sup_{[0, \gamma] \setminus \{\alpha\}} \delta(Dg, \alpha)} \right).$$

Notation: $\delta(Dg, 1)$ is the *difference quotient* of the function Dg w.r.t. a point 1:

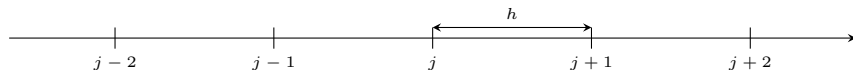
$$\delta(Dg, 1)(s) := \frac{(Dg)(1) - (Dg)(s)}{1 - s} = \frac{D(s)g(s)}{s - 1} > 0.$$

- Conditions (A)-(B)-(C) are a consequence of the fine estimates on proper sub-threshold speeds appearing in the study of solvability of the proper first-order problems.
- For any fixed D and g (regular enough), if f is sufficiently strictly concave then it satisfies the proper conditions in Theorem 2.
- The convexity assumption here has the role to better present the sufficient conditions but they **can be provided for a non-convex f** .

Applications:
a biological model with biased movements

-) *lattice-based stochastic model*, (see Johnston et al (2017) and Othmer-Stevenson (1997));

- agents are split in isolated and grouped individuals non-interacting;
- both classes can reproduce, die and move, with different rates;
- discrete model both in space $h > 0$ and time $\tau > 0$;



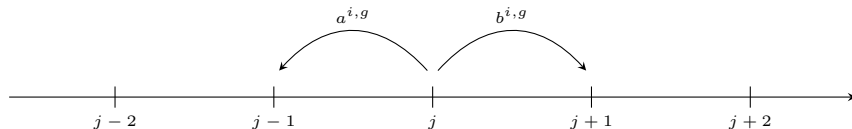
- c_j : probability of occupancy of the j -th site ($1 - c_j$: probability that j is vacant);
- P_m^i, P_m^g : “local” movement probabilities of (isolated and grouped) individuals;
- P_b^i, P_b^g : probabilities of birth;
- P_d^i, P_d^g : probabilities of death.

New features in **B.-Corli-Malaguti (2023)**:

Parameters $a^i, b^i \geq 0$ and $a^g, b^g \geq 0$

characterizing a **biased movement** (to the left and to the right). It could be expressed by probabilities.

-) Motivations: search for food, migration, escape from predators, ...
-) For example, isolated individuals move
 - to the left if $a^i - b^i > 0$;
 - to the right if $a^i - b^i < 0$.



-) Isolated agents at j have vacant sites on the two sides, grouped agents have at least one occupied site on a side.

-) The variation δc_j of c_j during a time-step $\tau > 0$ is

$$\delta c_j = \text{sum of some probabilities}$$

= an isolated at $j - 1$ move to the right +
 an isolated at $j + 1$ move to the left - an isolated move from $j +$ etc...

$$= P_m^i \left\{ b^i c_{j-1} (1 - c_j) (1 - c_{j-2}) + a^i c_{j+1} (1 - c_j) (1 - c_{j+2}) \right. \\ \left. - (a^i + b^i) c_j (1 - c_{j-1}) (1 - c_{j+1}) \right\} + \text{etc...}$$

-) $c_{j-1} (1 - c_j) (1 - c_{j-2})$: probability that site $j - 1$ is occupied but sites $j, j - 2$ are vacant.

The variation δc_j of c_j during a time-step $\tau > 0$ is

$$\begin{aligned} \delta c_j = & \frac{P_m^i}{2} \left[b^i c_{j-1} (1 - c_j) (1 - c_{j-2}) + a^i c_{j+1} (1 - c_j) (1 - c_{j+2}) \right. \\ & \left. - (a^i + b^i) c_j (1 - c_{j-1}) (1 - c_{j+1}) \right] \\ & + \frac{P_m^g}{2} \left[a^g c_{j-1} (1 - c_j) + b^g c_{j+1} (1 - c_j) - a^g c_j (1 - c_{j+1}) - b^g c_j (1 - c_{j-1}) \right] \\ & - \frac{P_m^g}{2} \left[a^g c_{j-1} (1 - c_j) (1 - c_{j-2}) + b^g c_{j+1} (1 - c_j) (1 - c_{j+2}) \right. \\ & \left. - (a^g + b^g) c_j (1 - c_{j-1}) (1 - c_{j+1}) \right] + \text{reaction terms}, \end{aligned}$$

where reaction terms =

$$\begin{aligned} = & \frac{P_b^i}{2} \left[c_{j-1} (1 - c_j) (1 - c_{j-2}) + c_{j+1} (1 - c_j) (1 - c_{j+2}) \right] \\ & + \frac{P_b^g}{2} \left[c_{j-1} (1 - c_j) + c_{j+1} (1 - c_j) \right] \\ & - \frac{P_b^g}{2} \left[c_{j-1} (1 - c_j) (1 - c_{j-2}) + c_{j+1} (1 - c_j) (1 - c_{j+2}) \right] - \frac{P_d^i}{2} \left[c_j (1 - c_{j-1}) (1 - c_{j+1}) \right] \\ & - \frac{P_d^g}{2} \left[c_j \right] + \frac{P_d^g}{2} \left[c_j (1 - c_{j-1}) (1 - c_{j+1}) \right]. \end{aligned}$$

The variation δc_j of c_j during a time-step $\tau > 0$ is

$$\begin{aligned} \delta c_j = & \frac{P_m^i}{2} \left[b^i c_{j-1} (1 - c_j) (1 - c_{j-2}) + a^i c_{j+1} (1 - c_j) (1 - c_{j+2}) \right. \\ & \left. - (a^i + b^i) c_j (1 - c_{j-1}) (1 - c_{j+1}) \right] \\ & + \frac{P_m^g}{2} \left[a^g c_{j-1} (1 - c_j) + b^g c_{j+1} (1 - c_j) - a^g c_j (1 - c_{j+1}) - b^g c_j (1 - c_{j-1}) \right] \\ & - \frac{P_m^g}{2} \left[a^g c_{j-1} (1 - c_j) (1 - c_{j-2}) + b^g c_{j+1} (1 - c_j) (1 - c_{j+2}) \right. \\ & \left. - (a^g + b^g) c_j (1 - c_{j-1}) (1 - c_{j+1}) \right] + \text{reaction terms}, \end{aligned}$$

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The continuum model is obtained as follows:

- replace c_j with a **smooth function** $c = c(x, t)$ and expand c around $x = jh$ at second order;
- divide by τ and **pass to the limit for $h, \tau \rightarrow 0$** while keeping h^2/τ constant (better, converging; for simplicity, think $h^2/\tau = 1$);
- **assume** for $l \rightarrow 0, \tau \rightarrow 0$:

$$\frac{P_m^{i,g} h^2}{2\tau} \sim D_{i,g}, \quad \frac{P_b^{i,g}}{2\tau} \sim \lambda_{i,g}, \quad \frac{P_d^{i,g}}{2\tau} \sim k_{i,g}, \quad P_b^{i,g}, P_d^{i,g} = O(\tau),$$

BIASED movements $\Rightarrow$

CONVECTION term in the continuum limit, as $h \rightarrow 0^+$.

($h > 0$ is the mesh size of the one-dimensional grid.)

$$a^{i,g}(\tau) \sim 1, \quad b^{i,g}(\tau) \sim 1, \quad a^{i,g}(\tau) - b^{i,g}(\tau) \sim C_{i,g} \sqrt{\tau}.$$

for some $C_{i,g} \begin{smallmatrix} \leq \\ \geq \end{smallmatrix} 0$.

In conclusion we end up with (u replaces c)

$$\boxed{u_t + f(u)_x = (D(u)u_x)_x + g(u)} \quad (M)$$

with

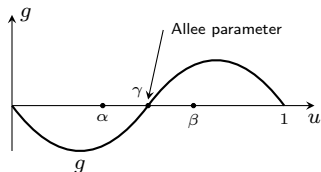
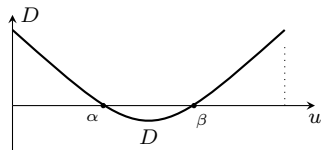
$$\begin{aligned} f(u) &= -(C_i D_i + C_g D_g) u(1-u)^2 - C_g D_g u(1-u), \\ D(u) &= D_i (1 - 4u + 3u^2) + D_g (4u - 3u^2), \\ g(u) &= \lambda_g u(1-u) + [\lambda_i - \lambda_g - (k_i - k_g)] u(1-u)^2 - k_g u. \end{aligned}$$

Eight parameters: $C_{i,g}$, $D_{i,g}$, $k_{i,g}$ and $\lambda_{i,g}$.

- $f(0) = f(1) = 0$: no convection at minimum (0) or maximum (1) density.
 f can change convexity at most once; then, it can be:

concave, convex, convex-concave, concave-convex.

FOCUS ON A PARTICULAR CASE: D VANISHES TWICE AND g IS BISTABLE



- D is as above iff $D_i > 4D_g > 0$; in this case

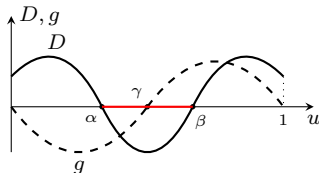
$$\alpha = \frac{2}{3} - \frac{\omega}{3} \quad \text{and} \quad \beta = \frac{2}{3} + \frac{\omega}{3}, \quad \text{for } \omega := \sqrt{\frac{D_i - 4D_g}{D_i - D_g}}.$$

- g is bistable as above iff $k_g = 0$ (think of a limited time horizon), $\lambda_g > 0$, $r_i := k_i - \lambda_i > 0$ (negative natural increase); in this case

$$\gamma = \frac{r_i}{r_i + \lambda_g},$$

and $\gamma \in (\alpha, \beta)$ if and only if $\frac{1-\omega}{2+\omega} < \frac{\lambda_g}{r_i} < \frac{1+\omega}{2-\omega}$.

RESULTS FOR D CHANGING SIGN TWICE: g BISTABLE



-) These diffusivities also occur in the modeling of crowds dynamics in overcrowded environments Colombo-Rosini (2005).
-) A previous result Kuzmin-Ruggerini (2011).

THEOREM

If $f = 0$ and a WF exists, then either $\gamma < \alpha$ and $c > 0$, or $\gamma > \beta$ and $c < 0$.

Then:

no WFs if $\gamma \in (\alpha, \beta)$.

THEOREM

$[f \neq 0]$ There exist c_1, c_0 such that:

-) If $c_1 < c_0$, then for every $c \in (c_1, c_0)$ there are WFs.

-) If $c_1 > c_0$, then no WFs.

- There are **infinitely many** profiles with the **same** speed $c \in (c_1, c_0)$.
- Explicit necessary conditions and sufficient conditions on D, g, f in order WFs exist.
- Conditions in order to have **positive** or **negative** speeds can be given.

- $C_i = 0 \Rightarrow$ no convective movement for the isolated individuals;
 f is convex if $C_g > 0$ and concave if $C_g < 0$.
- $C_g = 0 \Rightarrow$ no convective movement for the grouped individuals;
 f changes its concavity for $u = 2/3$.

PROPOSITION

Assume (M) admits wavefronts. Then

- $C_g < 0$;

THEOREM

- f concave $\Rightarrow$ there exists (according to the parameters r_i, λ_g) both WFs with positive speed (if d is large) and WFs with negative speed (if d is small).
- f convex $\Rightarrow$ no wavefronts.
- f convex-concave or concave-convex $\Rightarrow$ there exist wavefronts.

where

$$d := \frac{D_i}{D_g} > 4.$$

- Uniqueness of **smooth** profiles is lost in some cases; how to select a unique (meaningful) profile? (**entropy-like conditions**?)
- Prove **rigorously** the existence of **discontinuous** WFs. Since in this case

$$\left(\underbrace{D(u)}_{\text{discontinuous}} \cdot \underbrace{u_x}_{\sim \delta_x} \right)_x,$$

solutions are meant in a suitable sense, without the need of a regularization.
Caselles et al. (2015) in the context of Flux-saturated porous media equations

$$u_t + f(u)_x = \{D(u)\Phi(u_x)\}_x,$$

where $d, \Phi \in C^1$ and

$$d > 0 \text{ in } (0, 1), \quad d(0) = d(1) = 0;$$

$$\Phi : \mathbb{R} \rightarrow (-1, 1), \quad \Phi(\pm\infty) = \pm 1, \quad \Phi' > 0 \text{ in } \mathbb{R} \text{ (saturated flux)}.$$

An important example:

$$\Phi(u_x) = \frac{u_x}{\sqrt{1 + u_x^2}} \text{ (degenerate mean curvature operator).}$$

Thank you!